

Elements

The structure of the periodic table we use today was devised by Dmitri Mendeleev in 1869. It arranges all the elements in rows in order of increasing atomic number—the number of protons each atom has in its nucleus. Elements with similar properties and atomic structures are grouped together.

Periodic table

Every element is most easily identified by its symbol. The table has seven horizontal rows, called periods. The vertical columns are called groups. Periods 6 and 7 are too long to fit on the table, so the middle sections in Group 3 are shown at the bottom (see pp.188–189).

KEY

- Hydrogen
- Alkali metals
- Alkaline earth metals
- Transition metals
- Lanthanide series
- Actinide series
- Metalloids
- Other metals
- Other nonmetals
- Halogens
- Noble gases

List of elements

Atomic No	Name/Symbol	Discovery
1	hydrogen (H)	Henry Cavendish (1766)
2	helium (He)	William Ramsay (1895)
3	lithium (Li)	Johan Arfvedson (1817)
4	beryllium (Be)	Nicholas Louis Vauquelin (1797)
5	boron (B)	Louis-Josef Guy-Lussac; Louis-Josef Thénard; Humphry Davy (1808)
6	carbon (C)	Prehistoric
7	nitrogen (N)	Daniel Rutherford (1772)
8	oxygen (O)	Joseph Priestley; Carl Wilhelm Scheele (1774)
9	fluorine (F)	Henri Moissan (1886)
10	neon (Ne)	William Ramsay; Morris Travers (1898)
11	sodium (Na)	Humphry Davy (1807)
12	magnesium (Mg)	Joseph Black (1755)
13	aluminum (Al)	Hans Ørsted (1825)
14	silicon (Si)	Jöns Jacob Berzelius (1824)
15	phosphorus (P)	Hennig Brandt (1669)
16	sulfur (S)	Prehistoric
17	chlorine (Cl)	Carl Wilhelm Scheele (1774)
18	argon (Ar)	John Strutt; William Ramsay (1894)
19	potassium (K)	Humphry Davy (1807)
20	calcium (Ca)	Humphry Davy (1808)
21	scandium (Sc)	Lars Frederik Nilson (1879)
22	titanium (Ti)	William Gregor (1791)
23	vanadium (V)	Andrés Manuel del Rio (1801)
24	chromium (Cr)	Nicholas Louis Vauquelin (1798)
25	manganese (Mn)	Johan Gottlieb Gahn (1774)
26	iron (Fe)	Unknown (c 3500 BCE)

Atomic No	Name/Symbol	Discovery
27	cobalt (Co)	Georg Brandt (1739)
28	nickel (Ni)	Axel Cronstedt (1751)
29	copper (Cu)	Prehistoric
30	zinc (Zn)	Andreas Marggraf (1746)
31	gallium (Ga)	Paul-Émile Lecoq de Boisbaudran (1875)
32	germanium (Ge)	Clemens Winkler (1886)
33	arsenic (As)	Albertus Magnus (c 1250)
34	selenium (Se)	Jöns Jacob Berzelius (1817)
35	bromine (Br)	Antoine-Jérôme Balard; Carl Löwig (1826)
36	krypton (Kr)	William Ramsay; Morris Travers (1898)
37	rubidium (Rb)	Gustav Kirchhoff; Robert Bunsen (1861)
38	strontium (Sr)	Adair Crawford (1790)
39	yttrium (Y)	Johan Gadolin (1794)
40	zirconium (Zr)	Martin Heinrich Klaproth (1789)
41	niobium (Nb)	Charles Hatchett (1801)
42	molybdenum (Mo)	Peter Jacob Hjelm (1781)
43	technetium (Tc)	Carlo Perrier; Emilio Segrè (1937)
44	ruthenium (Ru)	Karl Karlovich Klaus (1844)
45	rhodium (Rh)	William Hyde Wollaston (1803)
46	palladium (Pd)	William Hyde Wollaston (1803)
47	silver (Ag)	Unknown (c 3000 BCE)
48	cadmium (Cd)	Friedrich Stromeyer (1817)
49	indium (In)	Ferdinand Reich; Hieronymous Richter (1863)
50	tin (Sn)	Unknown (c 2100 BCE)
51	antimony (Sb)	Unknown (c 1600 BCE)
52	tellurium (Te)	Franz-Joseph Müller von Reichenstein (1783)